Course Number: 125101 Course Name: Analytical Chemistry 1

Spring Semester 2024

Exam A

Student Name: _____

ID Number: _____

Program: _____

1. Bring your own **calculator** and **ID card** with you, and the hard dictionary will be provided by UG in the classroom;
2. Time allowed: 3 hours between 9:00-12:00 a.m.
3. All of the materials should be submitted;
4. Some important equations are attached in the **Appendix**;

Please read the instructions carefully:

- It includes ten multiple choice questions (MCQ) with only one correct answer and two open questions;
- Please briefly justify your choice, so that it is clear that it's not a random guess for your answer. Calculations: provide equations and details for the main steps;
- MCQ have just only **ONE** correct answer; No justification for the MCQ isn't acceptable; for open questions, the answers should be written down in the box of Answer Sheet.
- The maximum points that can be collected from the final exam is 100 (Each MCQ provides 6 points and two open questions provide 40 points.

Part One: multiple choice questions (Only **one correct** answer for each question, 60 point = 10×6 points)

Question No. 1 (6 pts.)

When you report the experimental results, which parameter is indispensable if you are interesting to the **measuring method**?

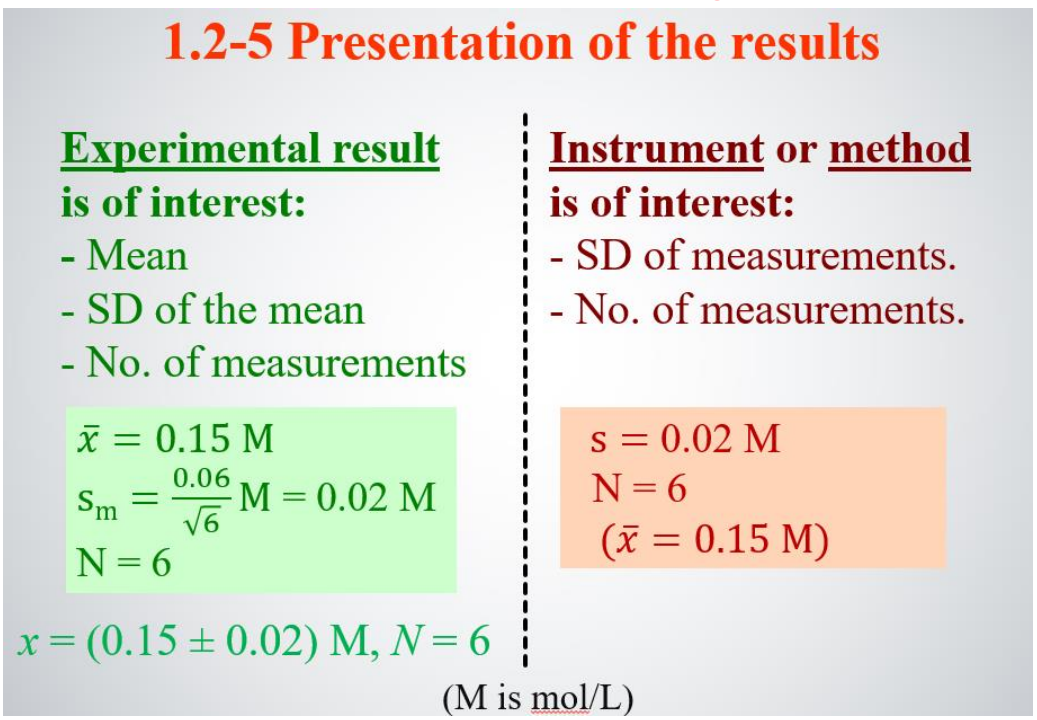
a. Mean of the measurements, $\bar{x}$

b. Standard deviation of the mean, s_m

c. Measurement number, n

d. How many groups for the tested sample

Justification: It includes s and n if the measuring method is interesting.



Question No. 2 (6 pts.)

A 100.00 mL sample of spring water was treated to convert any iron present to Fe^{2+}. Addition of 25.00 mL of 0.002517 M $K_2Cr_2O_7$ resulted in the reaction,

$$6Fe^{2+} + Cr_2O_7^{2-} + 14H^+ \Leftrightarrow 6Fe^{3+} + 2Cr^{3+} + 7H_2O$$

The excess $K_2Cr_2O_7$ was back-titrated with 8.53 mL of 0.01 M Fe^{2+} standard solution with the indicator. Calculate the concentration of iron in the sample in parts per million (ppm). [The molar mass of iron is about 56 g/mol]

a. 0.2 ppm of iron;

b. 2 ppm of iron;

c. 16 ppm of iron;

d. 164 ppm of iron;

Justification:

Total $K_2Cr_2O_7$ = 25.00 mL × 0.002517 mol/L = 0.063 mmol;

The left $K_2Cr_2O_7$ = $\frac{8.53\ mL \times 0.01\ mol/L}{6}$ = 0.014 mmol (in titration);

The reacted $K_2Cr_2O_7$ = Total $K_2Cr_2O_7$ - the left $K_2Cr_2O_7$ = 0.049 mmol (with Fe^{2+} in sample);

The Fe^{2+} in sample = 0.049 mmol × 6 = 0.294 mmol

The concentration of iron in the sample = $\frac{0.294\ mmol}{100.00\ mL}$ × 56 g/mol = 0.164 g/L

1 ppm = 1 μg/mL = 1 mg/L

The concentration of iron in the sample of 0.164 g/L= 164 mg/L = 164 ppm

Question No. 3 (6 pts.)

A student Yaniv was asked to evaluate the density of an unknown solution. He measured a solution 9 times and got average density of 1.8405 g/mL and the standard deviation was 0.0031 g/mL. According to the table Which of the following statement on the result is correct?

a. The measurement is about qualitative analysis;

b. The coefficient of variation (RSD) is 0.2%;

c. The standard deviation of 0.0031 g/mL belongs to systematic error;

d. The presentation of results should be, Density=1.841 (± 0.003) g/mL, N= 9;

Justification: a. The measurement is about quantitative (not qualitative)analysis;

c. The standard deviation of 0.0031 g/mL belongs to random (not systematic) error;

d. The presentation of results should be, Density=1.840 (± 0.001) g/mL, N= 9;

Question No. 4 (6 pts.)

In a weak acid titration with a strong base, the titration error due to the difference between the equivalent point and the titration endpoint is:

a. Is a random error that cannot be compensated for;

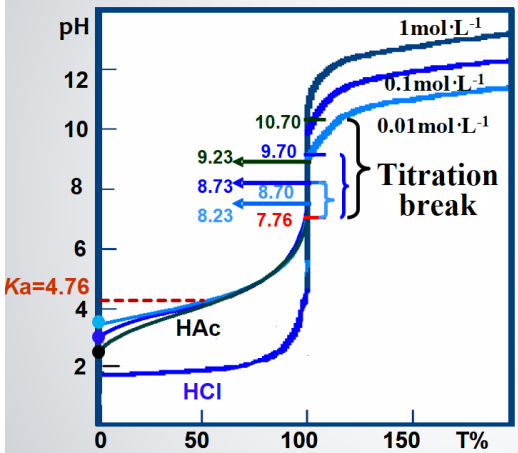
b. Decreases when reagents are more concentrated;

c. Increases when reagents are more concentrated;

c. Is independent on the reagents concentration, but depends on their number of moles;

Justification:

Titration curve with diff. [HAc]



$Ka(HAc) = 10^{-4.76}$

Titration break

0.01M: 7.76 – 8.70 by 0.01 M NaOH

0.10 M: 7.76 – 9.70 by 0.10 M NaOH

1.00 M: 7.76 – 10.70 by 1.00 M NaOH

For [HAc] increase 10 times, the titration break becomes larger of one pH unit

Titration break is larger at higher [C] of weak acid and titration error would be less.

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$$\text{Titration error} = \frac{C + C_0}{CC_0}\left[\frac{Kw}{[H^+]_{ep}} - [H^+]_{ep}\right] - \frac{[H^+]_{ep}}{Ka}$$

$$\text{Titration error} \approx \frac{C + C_0}{CC_0}\left[\frac{Kw}{[H^+]_{ep}} - [H^+]_{ep}\right] \quad \text{Strong acid}$$

Conclusions:

(a) One can calculate the titration error from the *p*H at end-point (indicator)

In titration of strong acid with strong base, using indicator at *p*H=7, error = 0

(b) Titration error increases upon reagent dilution

Given C, C_0:

$$\frac{C + C_0}{CC_0} = x$$

Dilution by a factor of 2:

$$\frac{\frac{1}{2}C + \frac{1}{2}C_0}{\frac{1}{2}C\,\frac{1}{2}C_0} = 2x$$

Question No. 5 (6 pts.)

What is the difference between methods between internal standard and external standard?

a. The standard compound of external standard is the same substance as analyte; However, it is different between them in internal standard;

b. In the calibration curve, *y* axis represents the height of analyte peak in external standard; and *y* axis represents the area of analyte peak in internal standard;

c. External standard was carried out when the signals are not reproducible or sample losses occur;

d. Internal standard can be used to address the matrix effect;

Justification:

Comparisons with calibration methods

Approaches	Comparison
External standard	Analyte with known [C] used to prepare a **calibration curve** to deduce the unknown
Standard addition	Added standard **is the same substance** as analyte to address matrix effect
Internal Standard	Added standard is **different from analyte** when the quantity of analyte is not reproducible or sample losses occur

Question No. 6 (6 pts.)

Based on what, potentiometry can be employed to determine the unknown in titrations?

a. Least squares;

b. Henderson equation;

c. Nernst equation;

d. Gran plot;

Justification:

The Nernst equation for the entire cell is:

$$E_{cell} = E_{+} - E_{-} = \underbrace{(0.799 - 0.05916 \log \frac{1}{[Ag^{+}]})}_{\text{Potential of Ag | Ag}^{+}\text{ indicator electrode}} - \underbrace{0.241}_{\text{Potential of S.C.E Reference electrode}}$$

$$\boxed{E_{cell} = 0.558 + 0.05916 \log [Ag^{+}]}$$

Question No. 7 (6 pts.)

In the following analytical methods, which one is involved with external standard?

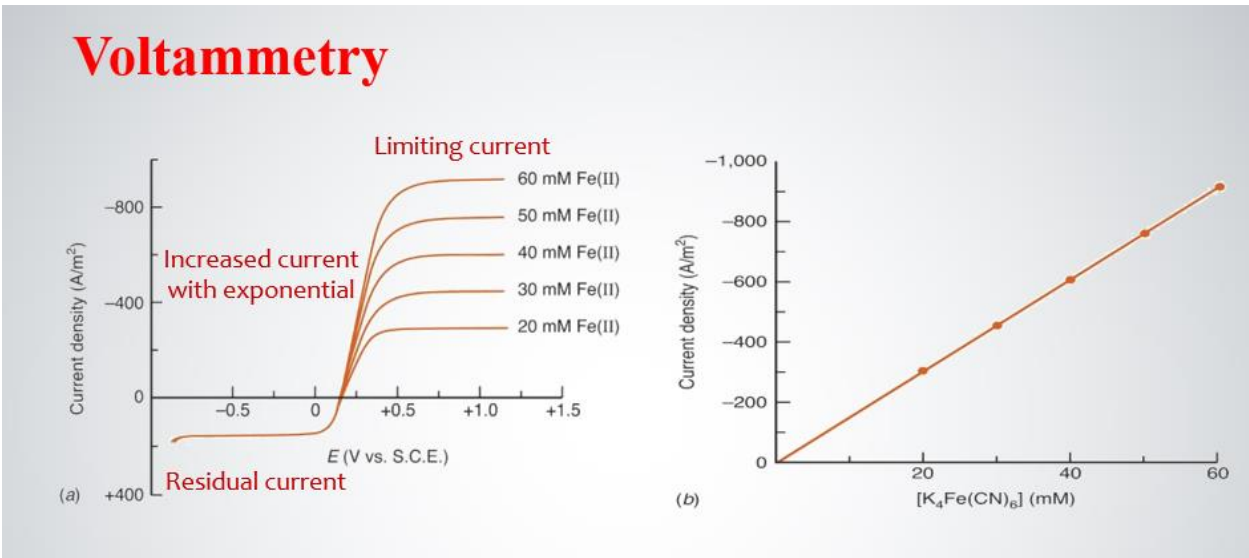
a. Gravimetric analysis;

b. Electrogravimetric analysis;

c. Coulometric analysis;

d. Voltammetry analysis;

Justification:



Question No. 8 (6 pts.)

In Voltammetry, the current as a function of the potential of the negative working electrode reaches saturation (after a certain increase). This saturation happens when…

a. The electrochemical reaction consumes all of the analyte in the solution;

b. The current is limited by the diffusion rate of the ions toward the electrode;

c. The voltage reaches the standard reduction potential of the analyzed ions;

d. The potential repeals negative ions from the electrode, thus the reaction stops;

Justification:

- Diffusion current is governed by the rate of analyte diffuses to working electrode surface.
- Current $\propto$ **diffusion rate** $\propto$ $(C_0 - C_s) \sim C_0$

"$\propto$" means "in proportion to"

Current is proportional to the C_0

This is the basis for quantitative analysis by Voltammetry

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Question No. 9 (6 pts.)

Calculate the potential required to initiate deposition of copper from a solution that is 0.010 M in $CuSO_4$ and contains sufficient H_2SO_4 to give a pH of 4.00.

[Note: The deposition of copper necessarily occurs at the cathode, which according to IUPAC convention is the right-hand-electrode. Since there is no more easily oxidizable species than water in the system, O_2 will evolve at the anode (the pressure of O_2 is assumed at 1 atm). The two half-reactions and their corresponding standard electrode potentials are followed,

$$Cu^{2+} + 2e- \leftrightarrow Cu(s) \; ; \; E° = +0.337 \text{ V}$$

$$O_2(g) + 4H^+ + 4e^- \leftrightarrow 2H_2O \; ; \; E° = +1.229 \text{ V}$$

a. A positive potential slightly greater than 0.714 V;

b. A negative potential slightly greater than 0.714 V;

c. A positive potential slightly less than 0.714 V;

d. A negative potential slightly less than 0.714 V;

Justification:

The net reaction is, $2Cu^{2+}(aq) + 2H_2O\,(l) \leftrightarrow 2Cu(s) + O_2(g) + 4H^+(aq)$

The $E = E^o(Cu^{2+}/Cu) + E^o(H_2O/H^+) - \frac{0.05916}{4}\log\frac{[H^+]^4 \cdot P(O2)}{[Cu^{2+}]^2}$

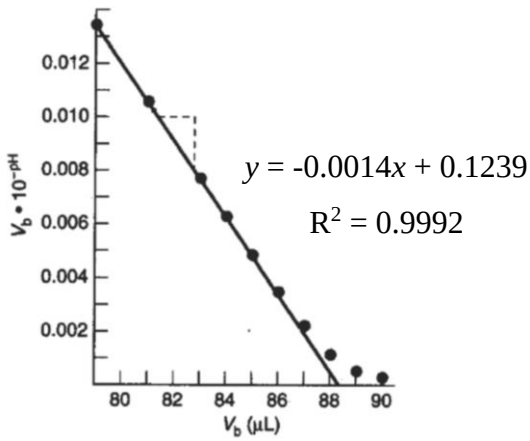
$= 0.337 \text{ V} - 1.229 \text{ V} - \frac{0.05916}{4}\log\frac{[10^{-4}]^4 \cdot P(1\ atm)}{[0.01]^2}$

$= -0.714 \text{ V}$

The potential required to initiate deposition of copper should be greater than -0.714 V (negative potential).

Question No. 10 (6 pts.)

In this experiment, you will titrate 1.0 mL of an unknown weak acid HA with standard 1.5 mol/L NaOH. To find end point of titration, the problem with using first derivative is that titration data are least accurate right near the end point because buffering is minimal and electrode response is sluggish. A Gran plot could be employed to locate the end point based on the data before the end point (typically from 0.8 V_e up to V_e). A graph of $V_b \cdot 10^{-pH}$ versus V_b is called a Gran plot, shown as below. Find the concentration and K_a of HA according to the Gran plot diagram.



a. [HA]= 0.13 mol/L; $K_a = 1.40\times10^{-3}$;

b. [HA]= 0.017 mol/L; $K_a = 1.35\times10^{-2}$;

c. [HA]= 0.017 mol/L; $K_a = 1.40\times10^{-3}$;

d. [HA]= 0.13 mol/L; $K_a = 1.35\times10^{-2}$;

Justification: Ve is the intercept of x axis. y= -0.0014x + 0.1239 = 0 ⟹ x = 88.5 μL

[HA]×1.0 mL = 1.5 mol/L NaOH]×Ve

$$\Longrightarrow [HA] = \frac{1.5\frac{mol}{L} \times 0.0885\ mL}{1\ mL} = 0.13 \text{ mol/L}$$

K_a is the absolute value of slope in the linear equation, K_a= -(-0.0014) = 1.4×10^{-3}

Part Two: open questions (40 point)

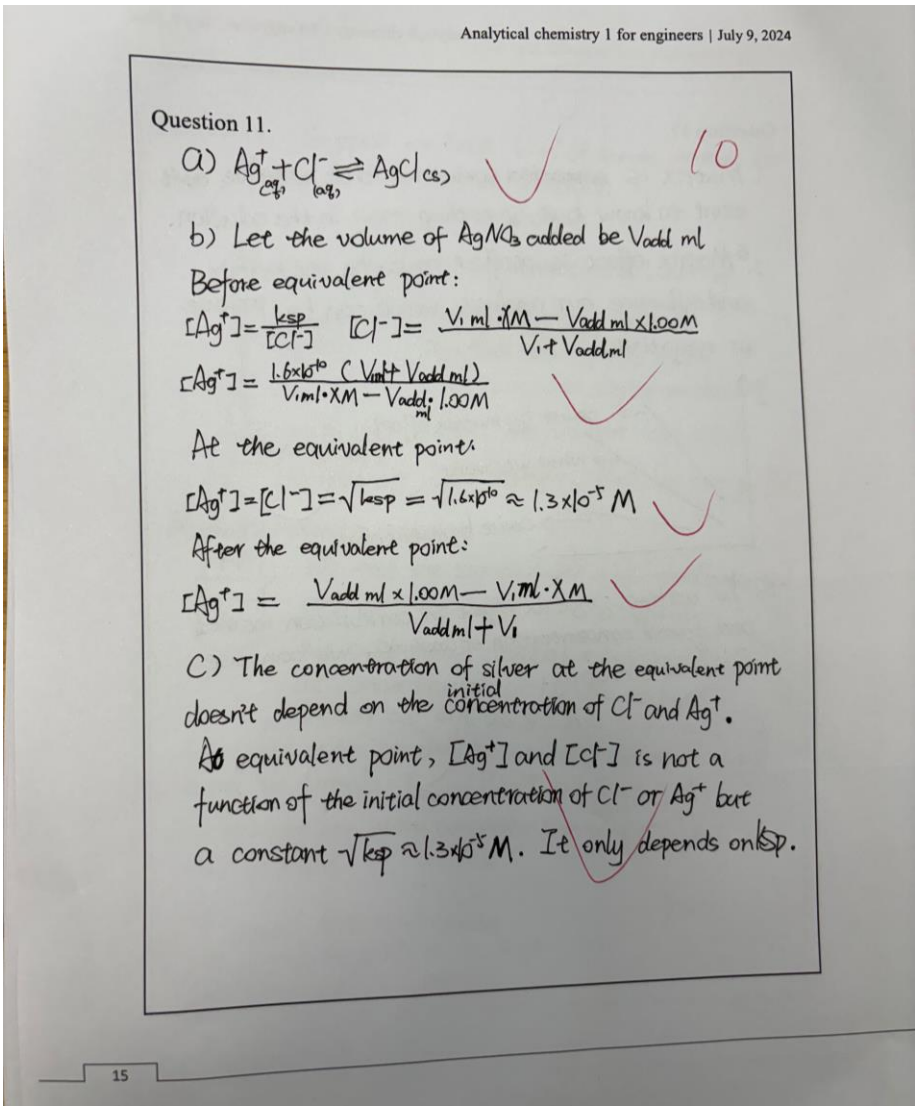
Question 11, Open question I (10 pts.)

In a precipitation titration of V_1 mL of a sample containing Cl^- at a concentration of *x* M using $AgNO_3$ at a concentration of 1.00 M as the titrant ($K_{sp} = 1.6 \times 10^{-10}$).

a). Define the titration reaction; (3 points)

b). Obtain the expressions to calculate the concentration of Ag^+ before, at and after the equivalent point as a function of the volume of $AgNO_3$ added; (4 points)

c). Does the concentration of silver at the equivalent point depends on the initial concentration of Cl^- and Ag^+ in the sample? Justify. (4 points)



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Question 11.

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a) $Ag^+_{(aq)} + Cl^-_{(aq)} \rightleftharpoons AgCl_{(s)}$

b) Let the volume of $AgNO_3$ added be V_{add} ml

Before equivalent point:

$[Ag^+] = \frac{k_{sp}}{[Cl^-]}$ $[Cl^-] = \frac{V_1 ml \cdot xM - V_{add} ml \times 1.00M}{V_1 + V_{add} ml}$

$[Ag^+] = \frac{1.6\times10^{-10} (V_1 ml + V_{add} ml)}{V_1 ml \cdot xM - V_{add} ml \cdot 1.00M}$

At the equivalent point:

$[Ag^+] = [Cl^-] = \sqrt{k_{sp}} = \sqrt{1.6\times10^{-10}} \approx 1.3\times10^{-5} M$

After the equivalent point:

$[Ag^+] = \frac{V_{add} ml \times 1.00M - V_1 ml \cdot xM}{V_{add} ml + V_1}$

c) The concentration of silver at the equivalent point doesn't depend on the initial concentration of Cl^- and Ag^+.

At equivalent point, $[Ag^+]$ and $[Cl^-]$ is not a function of the initial concentration of Cl^- or Ag^+ but a constant $\sqrt{k_{sp}} \approx 1.3\times10^{-5} M$. It only depends on k_{sp}.

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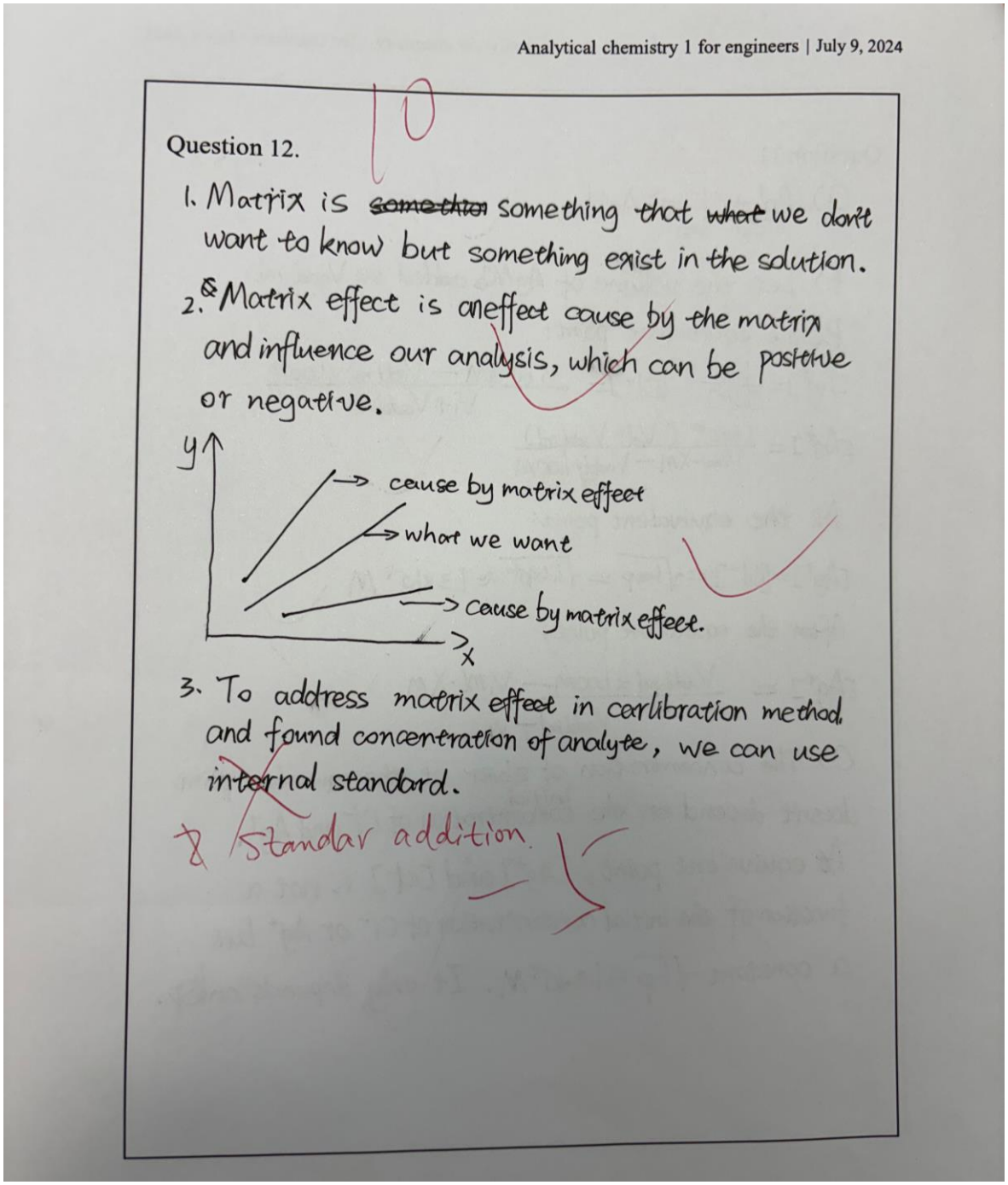
Question 12, Open question II (15 pts.)

What is matrix (5 points) and what is matrix effect? (5points) How to address the issue of matrix effect in calibration method and find the concentration of analyte? (5 points)



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Question 12.

1. Matrix is ~~somethin~~ something that ~~what~~ we don't want to know but something exist in the solution.

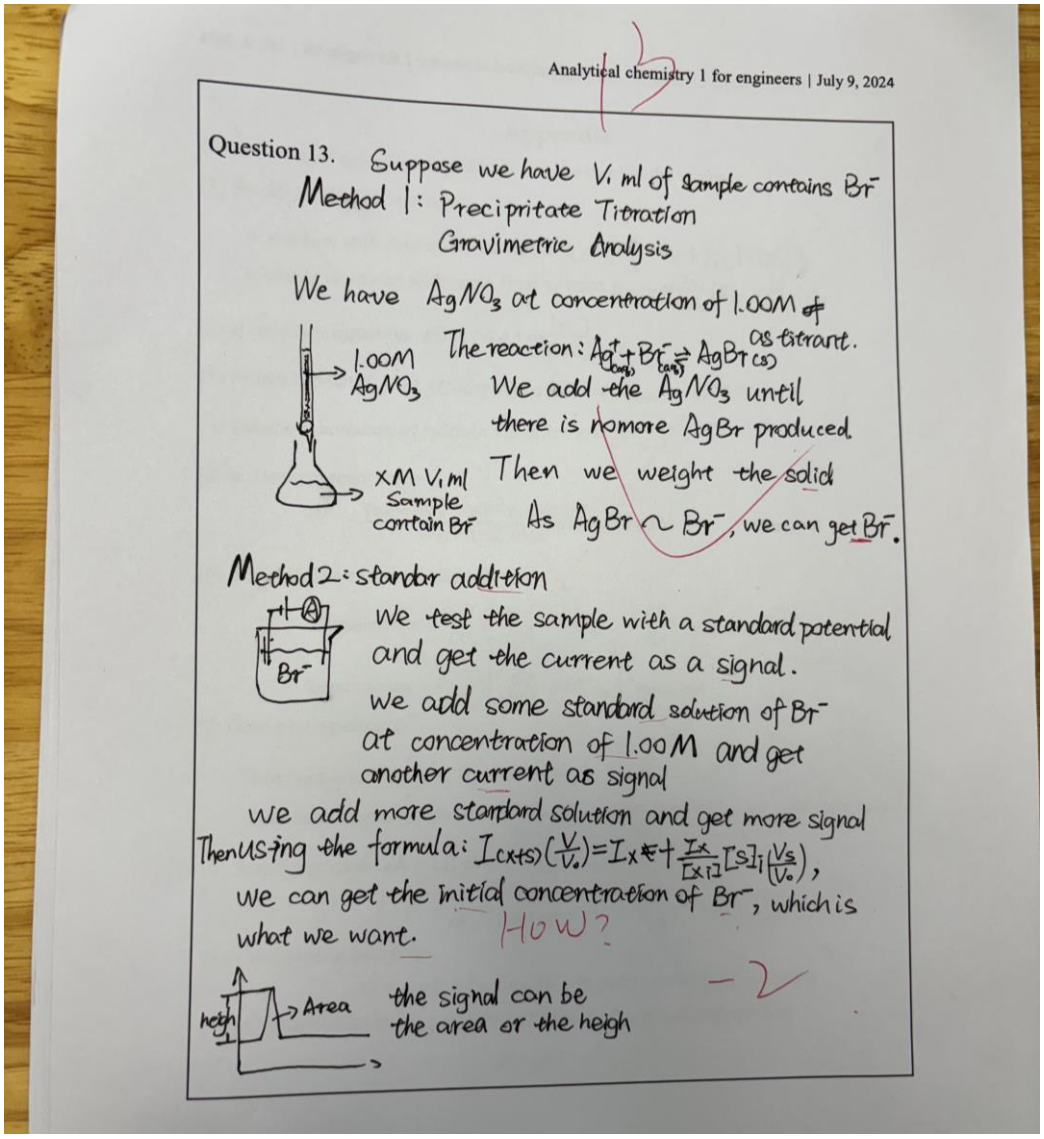
2. Matrix effect is aneffect cause by the matrix and influence our analysis, which can be postive or negative.

3. To address matrix effect in carlibration method and found concentration of analyte, we can use internal standard.

& Standar addition.

Question 13, Open question III (15 pts.)

Bromide ions (Br-) are present in many environments, such as sedative drugs, methyl-bromide-treated vegetables and seawater. Excess bromide in humans interferes with iodide metabolism and is considered toxic. The need for fast and inexpensive methods for bromide detection is of interest. Try to give at least two methods for the detection of Br^- in aqueous based on what you have learnt in the course. The principle, main steps, diagram and key parameters are supposed to be presented in your answer.



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Question 13. Suppose we have V_1 ml of sample contains Br^-

Method 1: Precipitate Titration
Gravimetric Analysis

We have $AgNO_3$ at concentration of 1.00M as titrant.

1.00M $AgNO_3$

xM V_1ml Sample contain Br^-

The reaction: $Ag^+_{(aq)} + Br^-_{(aq)} \rightleftharpoons AgBr_{(s)}$

We add the $AgNO_3$ until there is nomore AgBr produced

Then we weight the solid

As AgBr ~ Br^-, we can get Br^-.

Method 2: standar addition

Br^-

We test the sample with a standard potential and get the current as a signal.

we add some standard solution of Br^- at concentration of 1.00M and get another current as signal

we add more standard solution and get more signal

Then using the formula: $I_{(x+s)}(\frac{V}{V_0}) = I_x + \frac{I_x}{[x]_i}[S]_i(\frac{V_s}{V_0})$,

we can get the initial concentration of Br^-, which is what we want.

HOW?

-2

heigh

Area

the signal can be the area or the heigh

End of Exam!

Answer Sheet

Question items	Your answer
Question 1	**c**
Question 2	**d**
Question 3	**b**
Question 4	**b**
Question 5	**a**
Question 6	**c**
Question 7	**d**
Question 8	**b**
Question 9	**b**
Question 10	**a**

Question 11.

Question 12.

Question 13.

Appendix

Some formulas are following, which may be used in the exam.

(1) Standard addition method:

A solution with successive additions $I_{(X+S)}(\frac{V}{V_o})=I_X+\frac{I_X}{[X]_i}[S]_i(\frac{V_s}{V_o})$

Multiple solutions with same final volume $I_{(X+S)}=\frac{I_X}{[X]_i}[X]_f+\frac{I_X}{[X]_i}[S]_f$

(2) Henderson Equation: $pH = pKa + \log\frac{[A^-]}{[HA]}$

(3) Buffer capacity: $\beta = 2.3C\times\alpha(HA)\times\alpha(A^-) \Rightarrow \beta_{max.} = 0.575\ C$

(4) Standard deviation of random variables: $s=\sqrt{\frac{\sum_i(x_i-\bar{x})^2}{N-1}}$

(5) Relative percent error:

$$\frac{\text{Theoretical Yield} - \text{Actual Yield}}{\text{Theoretical Yield}}\times 100\%$$

(6) Titration error in Acid-base titration:

$$\text{Titration error} = \frac{C+C_0}{CC_0}\left[\frac{Kw}{[H^+]_{ep}}-[H^+]_{ep}\right]-\frac{[H^+]_{ep}}{Ka}\quad\text{Weak acid}$$

$$\text{Titration error} \approx \frac{C+C_0}{CC_0}\left[\frac{Kw}{[H^+]_{ep}}-[H^+]_{ep}\right]\quad\text{Strong acid}$$

(7) Gran plot equations:

For acid-base titration, $V_b\times10^{-pH} = K_aV_e - K_aV_b$

For redox titration, $V\times10^{-E/0.05916} = Ve\times10^{(Eref - Eo)/0.05916} - V\times10^{(Eref - Eo)/0.05916}$

(8) Nernst equation: aA + bB ⇌ cC + dD

$$E = E^\circ - \frac{RT}{nF}\ln\frac{[C]^c[D]^d}{[A]^a[B]^b} = E^\circ - \frac{0.059\ 16}{n}\log\frac{[C]^c[D]^d}{[A]^a[B]^b}$$

(9) Coulometric analysis:

$Q = n_e\times F = I_A\times t_{sec}$ ($F = 9.6485\times10^4$ C/mol; A=1C/s)

(10) Contaminants in the precipitate's liquid after *n* washing cycles:

$$X_n = X_0\left(\frac{u}{u+V/n}\right)^n$$